

Ass_Syn_2.0

You have to synthesize **your UART_TX** files using **Design Compiler shell** and generate **Technology Dependent** gate level netlists in Verilog format using standard cell libraries (typical, worst, best) inside **std_cells** folder:-

- **scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db**
- **scmetro_tsmc_cl013g_rvt_ff_1p32v_m40c**
- **scmetro_tsmc_cl013g_rvt_ss_1p08v_125c**

Required: -

1. Add the following Optimization Constraints in **cons.tcl**: -
 - Clock Frequency with 115.2 KHz
 - Clock uncertainty with 0.25 ns for setup
 - Clock uncertainty with 0.05 ns for hold
 - Clock transition with 0.1 ns
 - Input delays on all input ports except (CLK & RST) with 30% of clock period
 - Output delays on all output ports with 30% of clock period
 - Add Buffer driving cell for all input ports
 - Add load of 0.5 pf on all output ports
 - Add dont_touch attribute on CLK port
 - Set operation condition using slow and fast libraries
 - Set wireload model using slow library
2. Check the synthesis log (**syn.log**) is free of (**Errors, Latches, Comb Loops**) and modify your RTL files to fix any issues.
3. Generate the following reports: -
 - 1) Power report using (**report_power -hierarchy**) command
 - 2) Area report using (**report_area -hierarchy**) command
 - 3) Setup timing analysis report (**report_timing -max_paths 100 -delay_type max**) command
 - 4) Hold timing analysis report (**report_timing -max_paths 100 -delay_type min**) command
 - 5) Clocks report using (**report_clock -attributes**) command
 - 6) Constraints report using (**report_constraint -all_violators**) command

4. Generate the following Output files: -

- 1) Technology Dependent Gate Level Netlist in Verilog Format
- 2) Technology Dependent Gate Level Netlist in ddc Format
- 3) SDF File
- 4) SDC File

Required: -

A) You have to deliver the following files: -

- 0- Synthesis script >> syn_script.tcl
- 1- Constraints file >> cons.tcl
- 2- Synthesis log >> syn.log
- 3- Technology Dependent Verilog Netlist >> UART_TX.v
- 4- Technology Dependent ddc Netlist >> UART_TX.ddc
- 5- SDC File >> UART_TX.sdc
- 6- SDF File >> UART_TX.sdf
- 7- Area report >> Area.rpt
- 8- Power report >> power.rpt
- 9- Hold analysis report >> hold.rpt
- 10- Setup analysis report >> setup.rpt
- 11- Clocks report >> clocks.rpt
- 12- Constraints report >> constraints.rpt